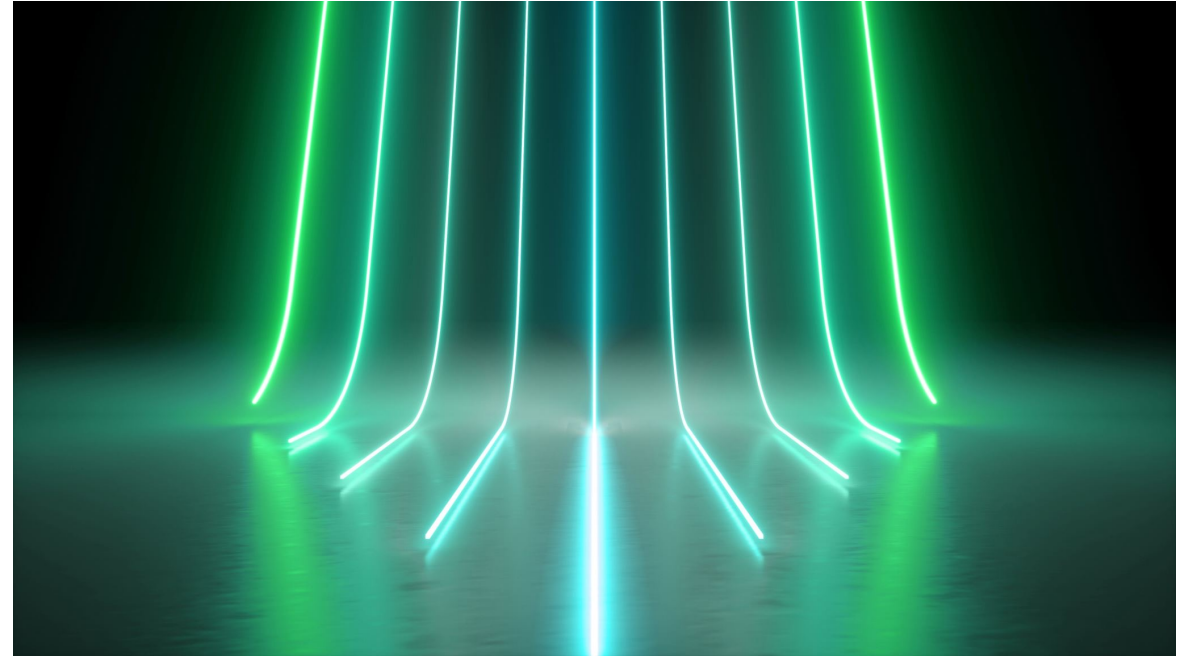


Continuous Random Variables

Stat 3113: Lesson 14



For Continuous Random Variable:

Probability Density Function (*pdf*):

The **probability density function** of a continuous random variable X is that function $f(x)$, with the following properties

1. $f(x) \geq 0$ for values of x

2. $\int_{-\infty}^{\infty} f(x) dx = 1$

3. $P(X \leq x) = \int_{-\infty}^x f(x) dx$

pmf versus pdf

- For a **discrete random** variable, we had a **probability mass function** (pmf).
- The pmf looked like a bunch of **spikes**, and probabilities were represented by the **heights of the spikes**.
- For a continuous random variable, we have a **probability density function** (pdf).
- The pdf looks like a **curve**, and probabilities are represented by **areas under the curve**.

Computing probability with the probability density function:

Let X be a continuous random variable with probability density function $f(x)$. Let a and b be any two numbers, with $a < b$. then

$$P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b) = P(a < X < b) = \int_a^b f(x) dx$$

In addition:

$$P(X \leq b) = P(X < b) = \int_{-\infty}^b f(x) dx$$

$$P(X \geq a) = P(X > a) = \int_a^{\infty} f(x) dx = 1 - P(X \leq a)$$

$$P(X = a) = P(a \leq X \leq a) = \int_a^a f(x)dx = 0$$

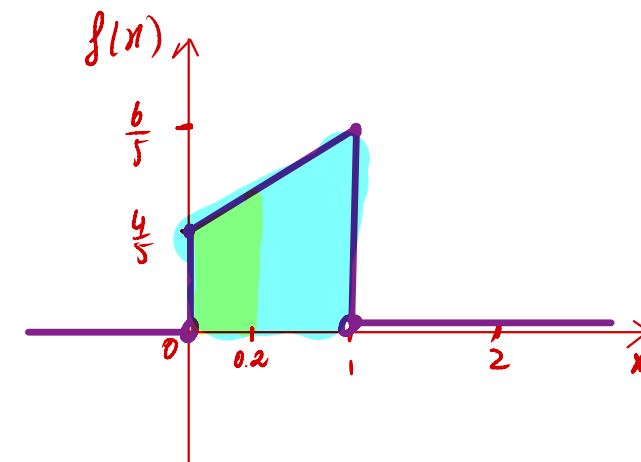
A continuous random variable has an infinite number of possible values & the probability of any one particular value is zero.

Example 1: The proportion of people who respond to a certain mail-order solicitation is a continuous random variable X that has the following probability density function:

$$f(x) = \begin{cases} \frac{2}{5}(x+2) & , 0 \leq x \leq 1 \\ 0 & , \text{ow} \end{cases}$$

a. Find $P(X \leq 0.2)$

$$\begin{aligned} P(X \leq 0.2) &= \int_{-\infty}^{0.2} f(x) dx = \underbrace{\int_{-\infty}^0 f(x) dx}_0 + \int_0^{0.2} f(x) dx \\ &= \int_0^{0.2} \frac{2}{5}(x+2) dx \approx 0.165 \end{aligned}$$



b. Find $P(0.5 < X \leq 0.75)$

$$\begin{aligned} P(0.5 < X \leq 0.75) &= \int_{0.5}^{0.75} f(x) dx = \int_{0.5}^{0.75} \frac{2}{5} (x+5) dx \\ &\approx 0.2625 \end{aligned}$$

c. Find $P(0.5 \leq X < 0.75) = P(0.5 < X \leq 0.75) \approx 0.2625$

c. Find $P(X > 0.65) = \int_{0.65}^1 \frac{2}{5} (x+2) dx$
 $\approx 0.3955.$

d. Find $P(X = 0.48) = \int_{0.48}^{0.48} f(x) dx = 0.$

Remark : you may use an online integral calculator. (link provided)

Expectation, Variance, and Std Deviation

- **Definitions**

- The **expected value** of a continuous random variable X is defined by

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

- Note the **similarity** to the definition for **discrete random variables**. Once again, we often denote it by μ . As in the discrete case this integral may not **converge**, in which case the expectation **if X is undefined**.

- Note that:

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

Expectation, Variance, and Std Deviation

- **Definitions**

- As in the discrete case we can use the **variance** via the **shortcut formula**

$$V(X) = \sigma^2 = \text{Var}(X) = E(X^2) - E(X)^2 = E(X^2) - \mu^2$$

- Once again the **standard deviation** is the **square root of variance**. Variance and standard deviation **do not exist if the expected value by which they are defined does not converge.**

$$\sigma = \sqrt{V(X)}$$

Example 1 (cont): The proportion of people who respond to a certain mail-order solicitation is a continuous random variable X that has the following probability density function:

$$f(x) = \begin{cases} \frac{2}{5}(x+2) & , 0 \leq x \leq 1 \\ 0 & , \text{otherwise} \end{cases}$$

a. Find $E(X)$ $= \int_{-\infty}^{\infty} x f(x) dx$ if this integral converges.

$$= \int_0^1 \frac{2}{5} x(x+2) dx \approx 0.5333$$

b. Find $V(X)$ and σ

$$\bullet V(X) = E[X^2] - \mu^2$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^1 \frac{2}{5} x^2 (x+2) dx \simeq 0.3667.$$

$$V(X) = 0.3667 - 0.5333^2$$

$$V(X) = 0.0823$$

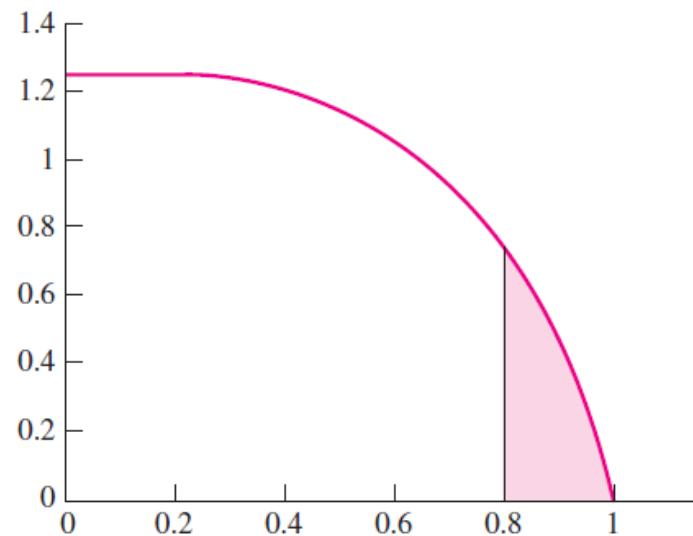
$$\bullet \sigma_X = \sqrt{V(X)} = \sqrt{0.0823} \simeq 0.28689.$$

Example 2:

A hole is drilled in a sheet-metal component, and then a shaft is inserted through the hole. The shaft clearance is equal to the difference between the radius of the hole and the radius of the shaft. Let the random variable X denote the clearance, in millimeters. The probability density function of X is

$$f(x) = \begin{cases} 1.25(1 - x^4) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Components with clearances larger than 0.8 mm must be scrapped. What proportion of components are scrapped?



$$\begin{aligned} P(X > 0.8) &= 1 - P(X \leq 0.8) \\ &= 1 - F(0.8) \\ &= 1 - \int_0^{0.8} 1.25(1 - x^4) dx \\ &\approx 0.08192. \end{aligned}$$

The population Median and Percentiles:

Let X be a continuous random variable with probability density function $f(x)$ and cumulative distribution function $F(x)$.

- The **median** of X is the point x_m that solve the equation:

$$F(x_m) = P(X \leq x_m) = \int_{-\infty}^{x_m} f(x) dx = 0.5 \quad (=\Rightarrow) \boxed{P(X > x_m) = 0.5}$$

- If p is any number between 0 and 100, the p^{th} **percentile** is the point x_p that solve the equation

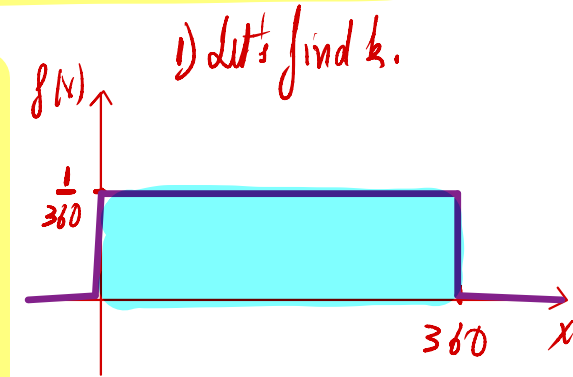
$$F(x_p) = P(X \leq x_p) = \int_{-\infty}^{x_p} f(x) dx = \frac{p}{100}$$

- The **median** is the 50th percentile.

Example 3: The direction of an **imperfection** with respect to a reference line on a circular object such as a tire, brake rotor, or flywheel is, in general, subject to uncertainty. Consider the reference line connecting the valve stem on a tire to center point, and let **X** be the angle measured clockwise to the location of an **imperfection**. One possible pdf for X is:

$$f(x) = \begin{cases} \frac{1}{k} & \text{for } 0 \leq x \leq 360 \\ 0 & \text{otherwise} \end{cases}$$

1. Find k .
2. Find $P(\text{the angle of occurrence is between } 90^\circ \text{ and } 180^\circ)$.
3. Find the probability that the angle of occurrence is above 90° .
4. Find the mean and standard deviation of X .
5. Find the median and the first quartile.



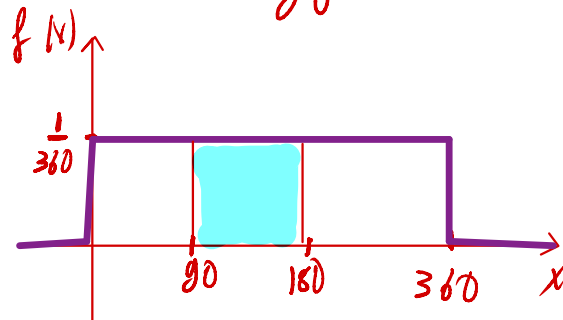
$$\begin{aligned} \int_0^{360} f(x) dx &= 1 \quad (\Rightarrow) \quad \int_0^{360} \frac{1}{k} dx = 1 \\ &(\Rightarrow) \quad \frac{1}{k} \int_0^{360} dx = 1 \\ &(\Rightarrow) \quad \frac{1}{k} [x]_0^{360} = 1 \\ &(\Rightarrow) \quad \frac{360}{k} = 1 \\ &(\Rightarrow) \quad \boxed{k = 360} \end{aligned}$$

Example 4: The direction of an imperfection with respect to a reference line on a circular object such as a tire, brake rotor, or flywheel is, in general, subject to uncertainty. Consider the reference line connecting the valve stem on a tire to center point, and let X be the angle measured clockwise to the location of an imperfection. One possible pdf for X is

$$f(x) = \begin{cases} \frac{1}{k} & \text{for } 0 \leq x \leq 360 \\ 0 & \text{otherwise} \end{cases}$$

2. Find $P(\text{the angle of occurrence is between } 90^\circ \text{ and } 180^\circ)$.

$$P(90 < X < 180) = \int_{90}^{180} \frac{1}{360} dx = \frac{1}{360} (180 - 90) = \frac{90}{360} = 0.25$$

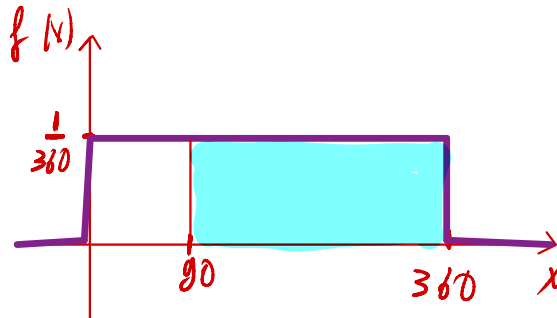


Example 5: The direction of an imperfection with respect to a reference line on a circular object such as a tire, brake rotor, or flywheel is, in general, subject to uncertainty. Consider the reference line connecting the valve stem on a tire to center point, and let X be the angle measured clockwise to the location of an imperfection. One possible pdf for X is

$$f(x) = \begin{cases} \frac{1}{k} & \text{for } 0 \leq x \leq 360 \\ 0 & \text{otherwise} \end{cases}$$

3. Find the probability that the angle of occurrence is above 90° .

$$P(X > 90) = \int_{90}^{360} \frac{1}{360} dx = \frac{1}{360} (360 - 90) = 0.75$$



Example 6: The direction of an imperfection with respect to a reference line on a circular object such as a tire, brake rotor, or flywheel is, in general, subject to uncertainty. Consider the reference line connecting the valve stem on a tire to center point, and let X be the angle measured clockwise to the location of an imperfection. One possible pdf for X is

$$f(x) = \begin{cases} \frac{1}{k} & \text{for } 0 \leq x \leq 360 \\ 0 & \text{otherwise} \end{cases}$$

4. Find the mean and standard deviation of X .

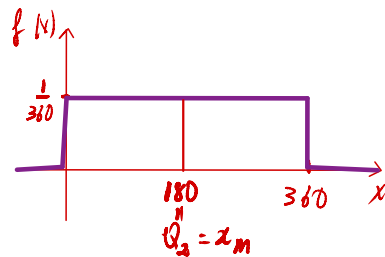
$$\begin{aligned} \bullet \ E[X] &= \int_0^{360} \frac{x}{360} dx = \frac{1}{360} \int_0^{360} x dx = \frac{1}{360} \left[\frac{1}{2} x^2 \right]_0^{360} = \frac{1}{360} \left(\frac{1}{2} (360)^2 \right) = 180. \\ \bullet \ E[X^2] &= \int_0^{360} \frac{x^2}{360} dx = \frac{1}{360} \int_0^{360} x^2 dx = \frac{1}{360} \left[\frac{1}{3} x^3 \right]_0^{360} = \frac{1}{360} \left(\frac{1}{3} (360)^3 \right) = 43200. \\ \bullet \ V(X) &= E[X^2] - \mu^2 = 43200 - (180)^2 = 10800 \Rightarrow \sigma_X = \sqrt{10800} = 60\sqrt{3}. \end{aligned}$$

Example 7: The direction of an imperfection with respect to a reference line on a circular object such as a tire, brake rotor, or flywheel is, in general, subject to uncertainty. Consider the reference line connecting the valve stem on a tire to center point, and let X be the angle measured clockwise to the location of an imperfection. One possible pdf for X is

$$f(x) = \begin{cases} \frac{1}{k} & \text{for } 0 \leq x \leq 360 \\ 0 & \text{otherwise} \end{cases}$$

5. Find the median and the first quartile.

• Median:



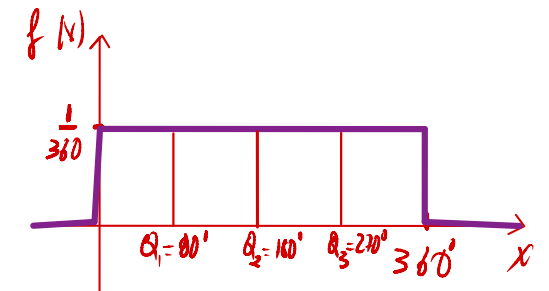
$$\begin{aligned} IP(X \leq x_m) &= 0.5 \\ \Rightarrow \int_0^{x_m} \frac{1}{360} dx &= 0.5 \\ \Rightarrow \frac{x_m}{360} &= 0.5 \\ \Rightarrow x_m &= \frac{360}{2} = 180. \end{aligned}$$

• 1st quantile = Q_1

$$IP(X \leq x_{25}) = 0.25$$

$$\Rightarrow \int_0^{x_{25}} \frac{1}{360} dx = 0.25$$

$$\Rightarrow \frac{x_{25}}{360} = 0.25 \Rightarrow x_{25} = \frac{360}{4} = 90.$$



Reading

Chapter 3 – Section 3.3

20

Next Topic



Cumulative Distribution Function for
Continuous Random Variables



Chapter 3 - Section 3.3